LG Patrols Effort Workflow — User Guide

This guide walks you through configuring and running the LG Patrols Effort Workflow, which generates a patrol effort analysis report for Lion Guardians rangers in the Amboseli ecosystem sourced from EarthRanger.

Overview

The workflow produces, for each patrol group:

- A **Patrol Events map** (scatter plot of events by type)
- A **Patrol Trajectories map** (tracks coloured by a user-selected category)
- A **Linear Time Density map** (patrol coverage heat map)
- A **time series bar chart** and **pie chart** of patrol events
- **Summary CSV tables** — per guardian, patrol type, event type, and monthly breakdown
- A **Word document report** (`.docx`) with a cover page and one section per group
- An **interactive widget dashboard**

Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance with a configured data source
- The patrol types and event types you want to analyse (leave blank to include all)

> The three spatial boundary files (group ranch boundaries, conflict hotspot areas, and protected areas) are downloaded automatically from Dropbox — no local copies are required.

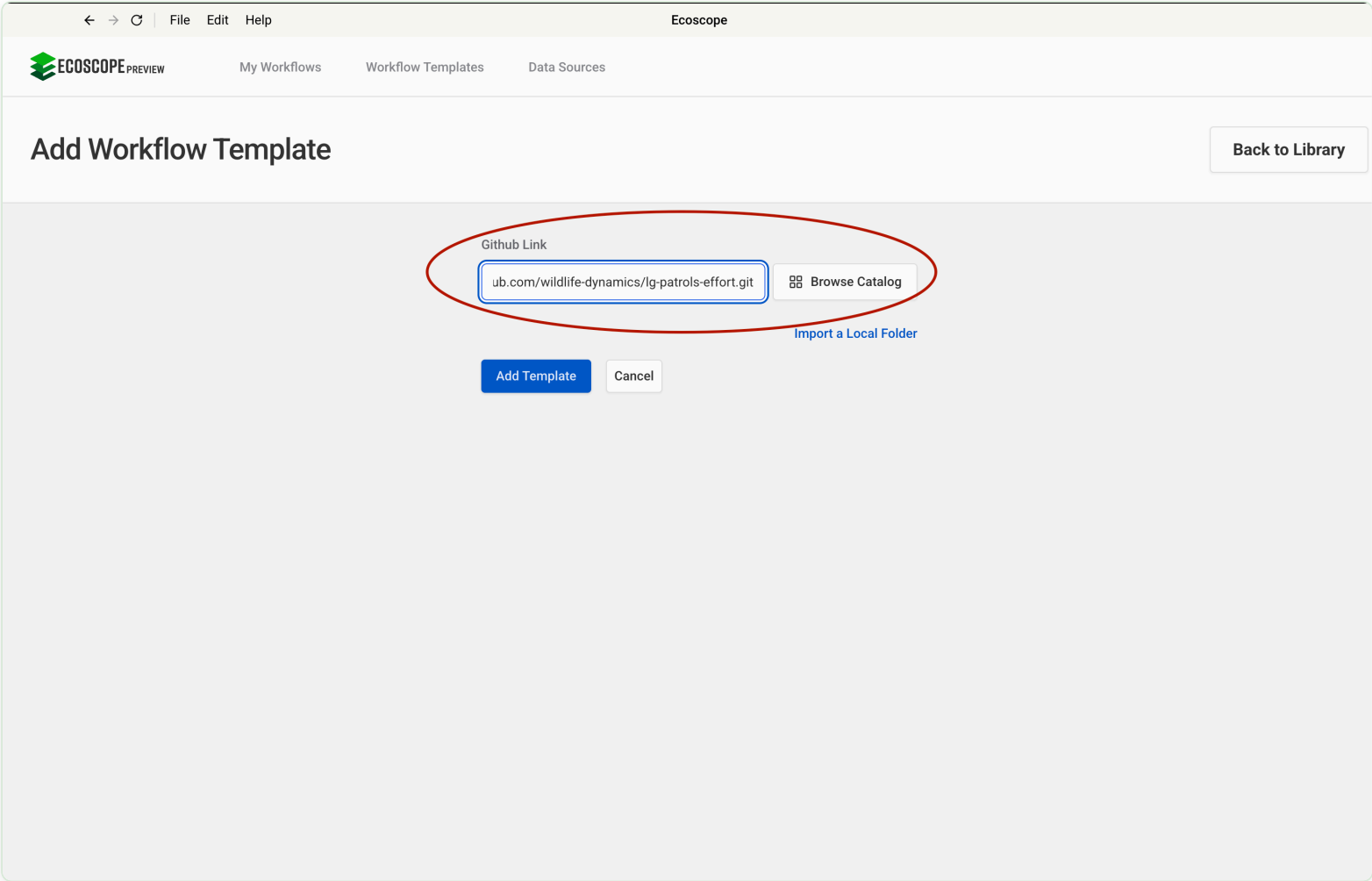
Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field:

```
https://github.com/wildlife-dynamics/lg-patrols-effort.git
```

Then click **Add Template**.



Add Workflow Template

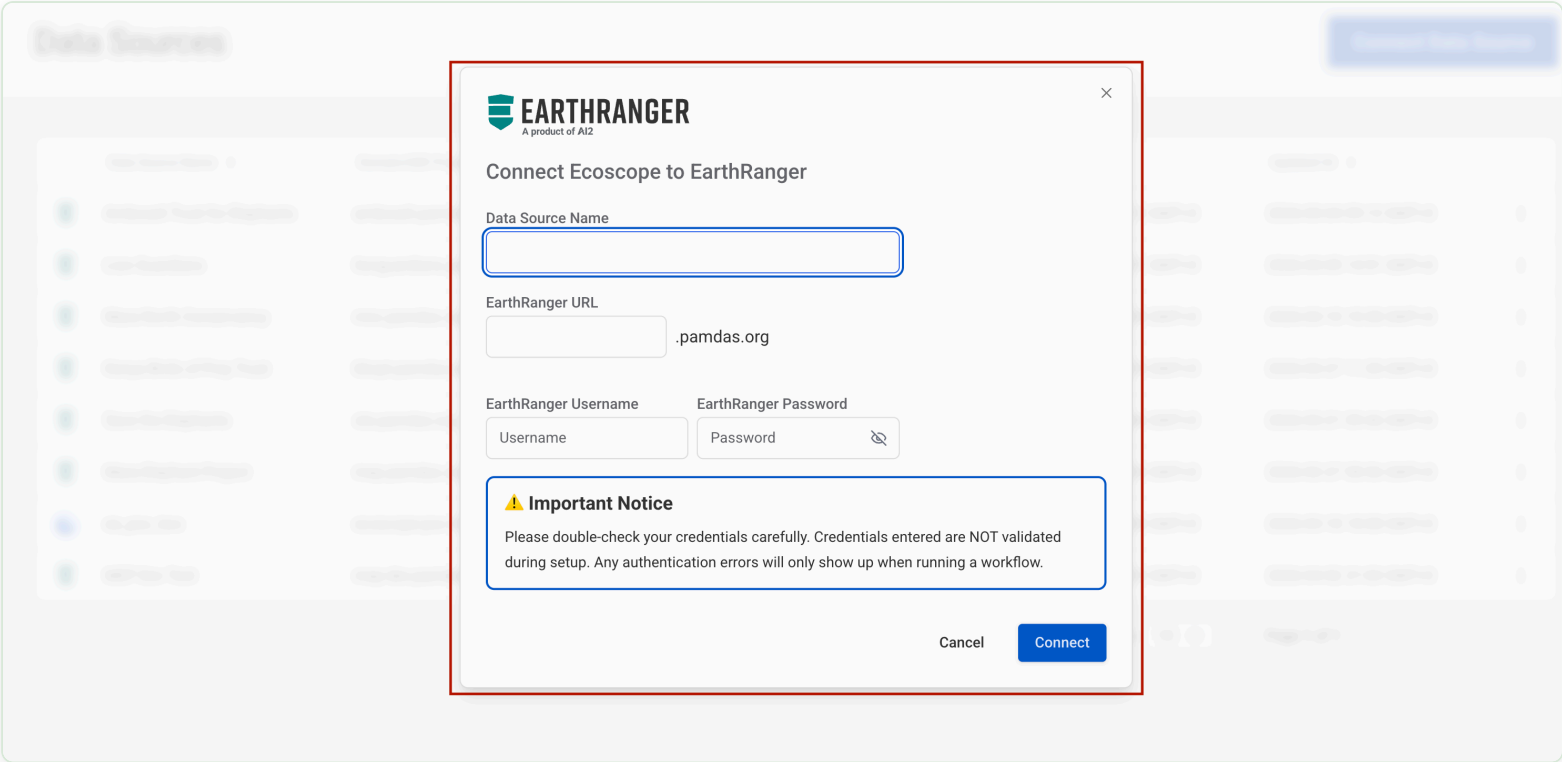
Step 2 — Add an EarthRanger Connection

Navigate to **Data Sources** and add a new EarthRanger connection. Fill in:

- **Data Source Name** — a label to identify this connection
- **EarthRanger URL** — your instance URL (e.g. `your-site.pamdas.org`)

- **EarthRanger Username** and **EarthRanger Password**

> Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

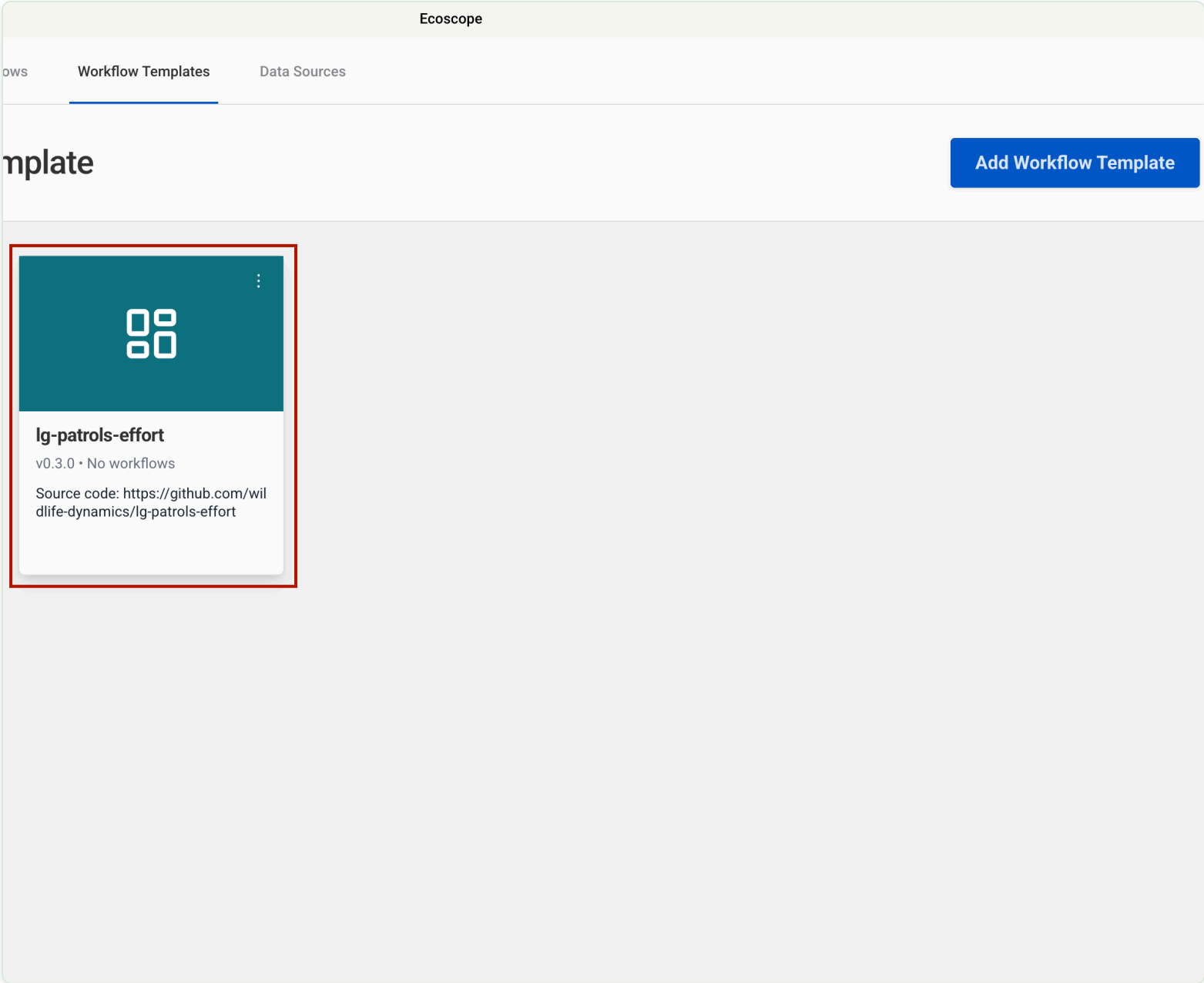


EarthRanger Connection

Step 3 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **lg-patrols-effort**. Click it to open the workflow configuration form.

> The card may show **Initializing...** briefly while the environment is set up.



Select Workflow Template

Step 4 — Configure Workflow Details and Time Range

The configuration form opens with two sections at the top.

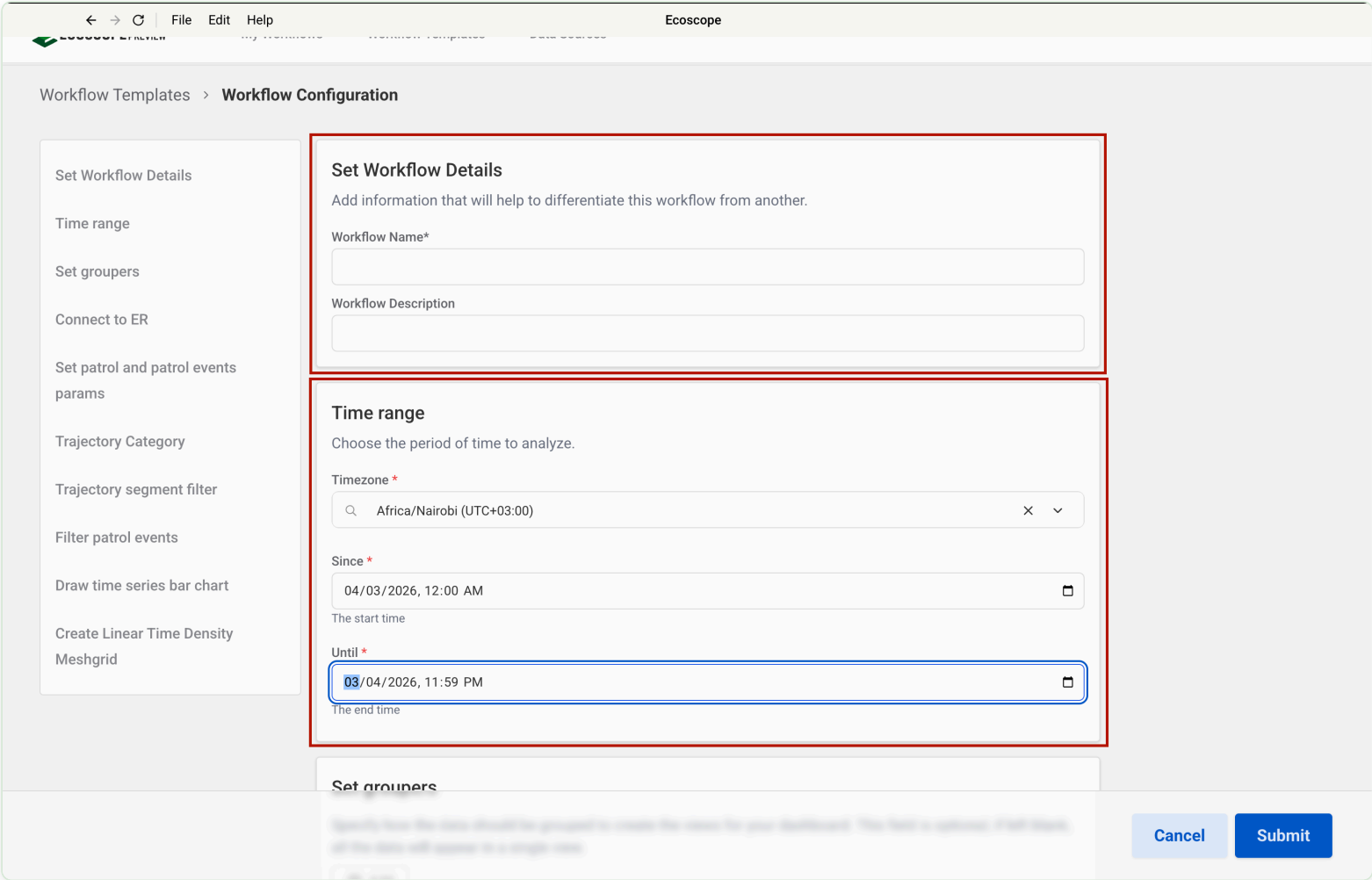
Set Workflow Details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes about the run (e.g. date range or patrol group)

Time range

Field	Description
Timezone	Select the local timezone (e.g. `Africa/Nairobi UTC+03:00`)
Since	Start date and time of the analysis period
Until	End date and time of the analysis period

All patrol tracks, events, and metrics are computed within this window.



Configure Workflow Details and Time Range

Step 5 — Set Groupers, Connect to ER, and Set Patrol Parameters

Scroll down to configure three sections.

Set Groupers *(optional)*

Groupers control how the workflow partitions data for per-group outputs. If left blank, all data appears in a single view. Click **Add** to add a grouper. Available options:

Grouper	Effect
Patrol Type	One map, metrics, and report section per patrol type
Patrol Serial Number	One output per patrol serial
Patrol Subject	One output per guardian ranger

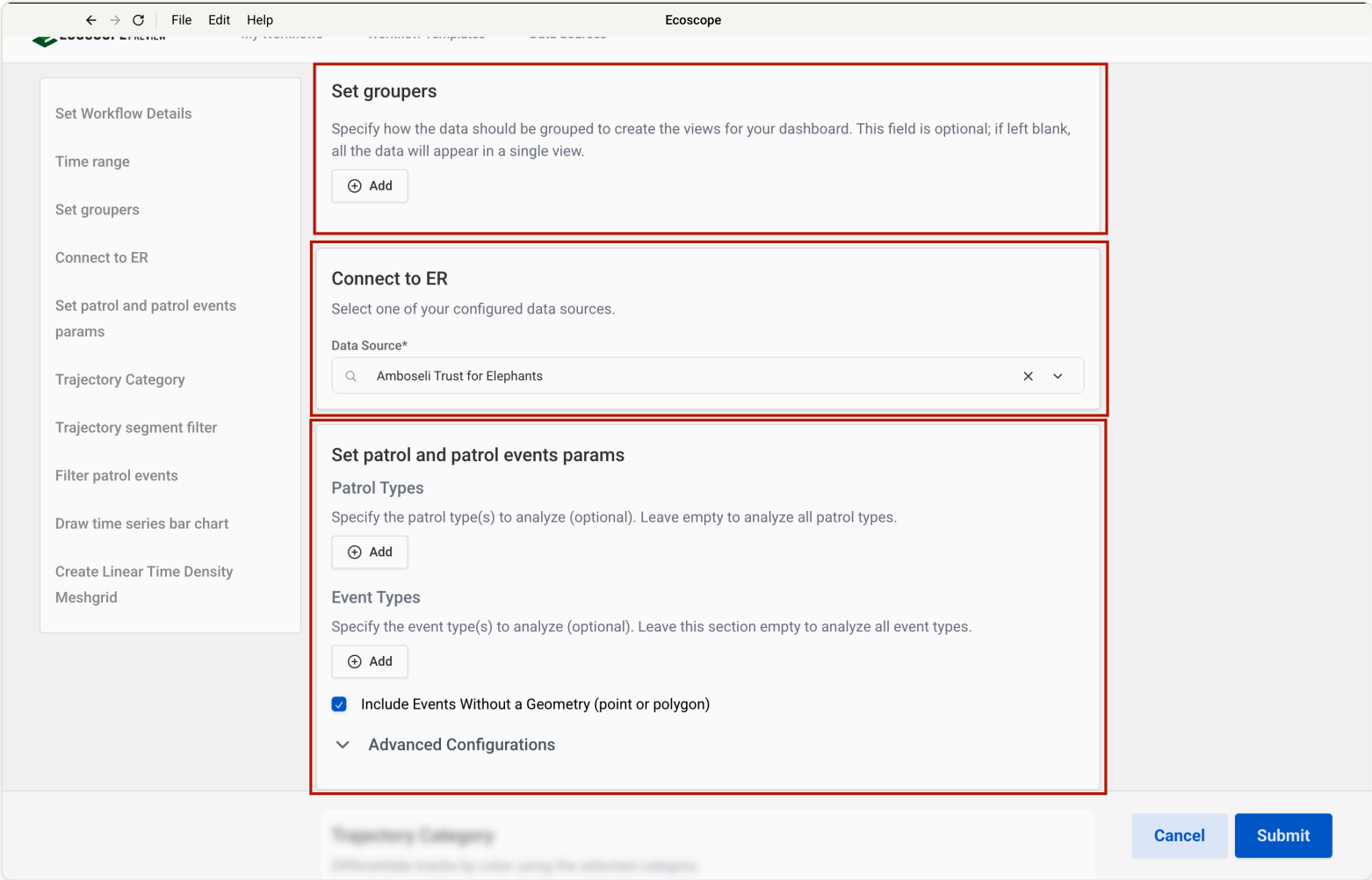
Connect to ER

Select the EarthRanger data source configured in Step 2 from the **Data Source** dropdown.

Set patrol and patrol events params

Field	Description
Patrol Types	Filter to specific patrol types (leave empty to include all)
Event Types	Filter to specific event types (leave empty to include all)
Include Events Without a Geometry	Check to include events with no point or polygon location

Expand **Advanced Configurations** to access additional query parameters such as patrol status and overlap behaviour.



Set Groupers, Connect to ER, and Patrol Parameters

Step 6 — Trajectory Category, Segment Filter, Event Filter, Bar Chart, and Time Density

The final section of the form contains five configuration panels.

Trajectory Category

Select the column used to colour patrol tracks on the Trajectories map:

Option	Effect
Patrol Type	Tracks coloured by patrol activity type
Patrol Subject	Tracks coloured per ranger
Patrol Serial Number	Tracks coloured per individual patrol

Trajectory segment filter *(Advanced Configurations)*

These parameters remove GPS noise and biologically unrealistic movements before trajectory analysis.

Field	Default	Description
Minimum Segment Length (m)	`0.001`	Discard segments shorter than this distance
Maximum Segment Length (m)	`100000`	Discard segments longer than this distance
Minimum Segment Duration (s)	`1`	Discard segments shorter than this duration
Maximum Segment Duration (s)	`172800`	Discard segments longer than this duration (48 hours)
Minimum Segment Speed (km/h)	`0.01`	Discard segments below this average speed
Maximum Segment Speed (km/h)	`500`	Discard segments above this average speed

Filter patrol events *(Advanced Configurations)*

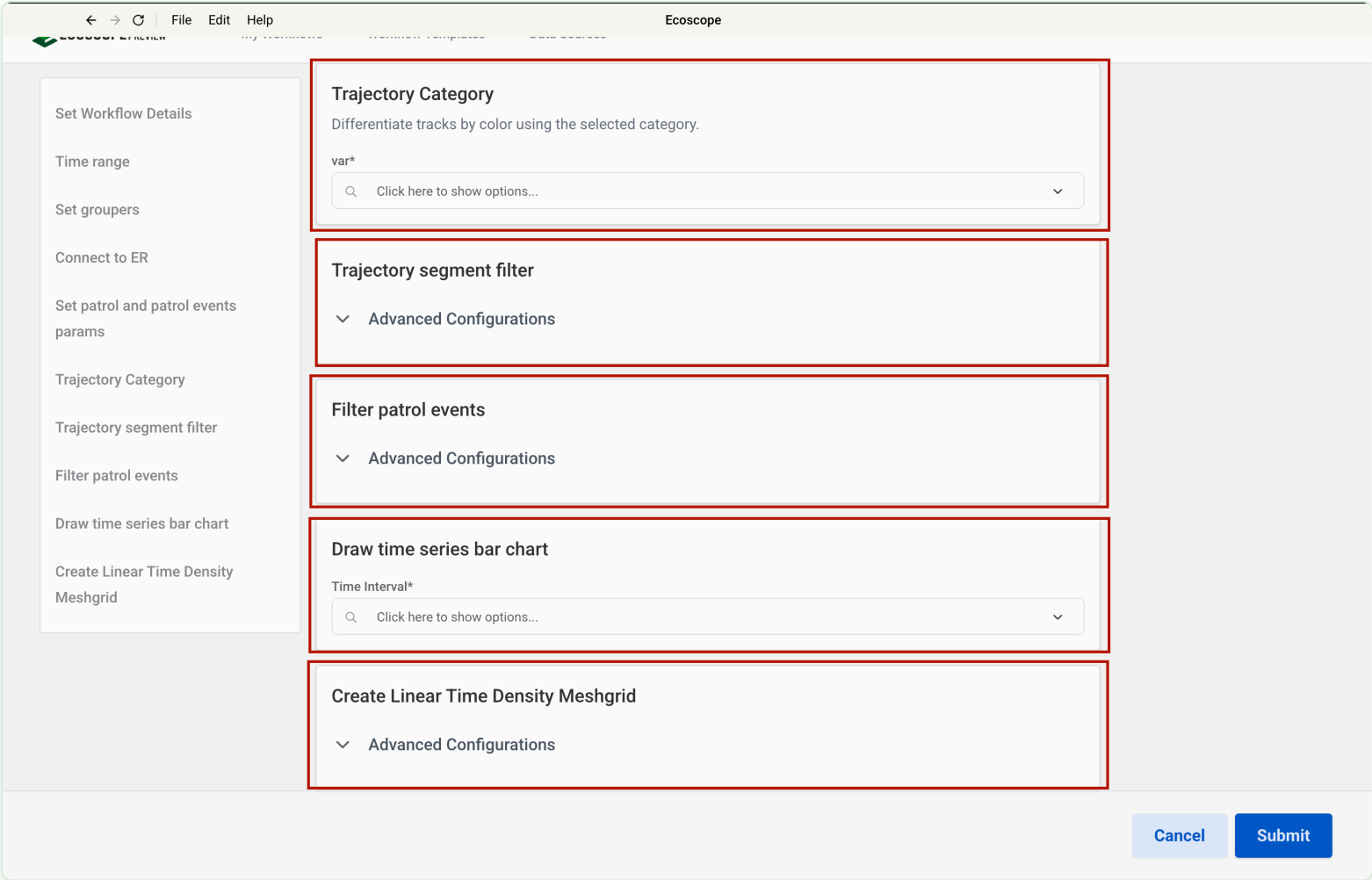
Optionally restrict events to a region of interest (ROI). Leave blank to include all events within the time range.

Draw time series bar chart

Field	Description
Time Interval	The time bucket used to group events on the x-axis (e.g. day, week, month)

Create Linear Time Density Meshgrid *(Advanced Configurations)*

Controls the resolution of the patrol coverage raster. Leave at defaults for most analyses.



Trajectory Category, Segment Filter, Bar Chart, and Time Density

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Running the Workflow

Once all parameters are configured, click **Submit**. The runner will:

1. Pull patrol tracks and events from EarthRanger for the specified time range.

2. Download the static boundary files (group ranches, conflict hotspots, protected areas).
3. Convert observations to relocations and build trajectory segments.
4. Generate the Patrol Events, Patrol Trajectories, and Linear Time Density maps.
5. Compute per-guardian, patrol type, event type, and monthly summary tables.
6. Render the time series bar chart and pie chart.
7. Assemble the Word report (cover page + per-group sections) and the dashboard.
8. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`:

File	Description
`events.geoparquet`	Patrol events with display names and timezone-converted timestamps
`trajectories.geoparquet`	Patrol segments with speed and distance
`<group>_events.html`	Interactive patrol events map per group
`<group>_patrol_trajectories.html`	Interactive patrol trajectories map per group
`<group>_time_density.html`	Interactive linear time density map per group
`<group>_events.png`	Screenshot of the patrol events map (2× resolution)
`<group>_time_density.png`	Screenshot of the time density map (2× resolution)
`<group>_patrols_pie_chart.png`	Screenshot of the event type pie chart (2× resolution)
`<group>_bar_chart.png`	Screenshot of the events time series bar chart (2× resolution)

File	Description
`<group>_guardian_patrol.csv`	Per-guardian patrol effort (patrols, distance, time)
`<group>_guardian_events.csv`	Per-guardian event count
`<group>_pivot_guardian_events.csv`	Guardian × event type pivot table
`<group>_monthly_patrol_efforts.csv`	Monthly patrol effort summary
`<group>_event_types.csv`	Per-event-type count
`context_page.docx`	Rendered report cover page
`<group>.docx`	Per-grouper report section
Merged report `.docx`	Final combined Word report